Math Kangaroo Practice - Grades 3-4

Practice set for 2002 #10, 2002 #11, 2002 #12
Each original problem is followed by five similar practice problems.

Problem 1: 2002 #10 - Digit Occurrences in House Numbers

Counting how many times a digit appears across a sequential range of numbers

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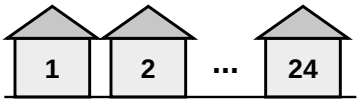
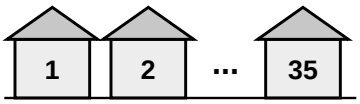
2002 #10 Joseph lives on a short street where the houses are numbered sequentially from 1 to 24. How many times does the digit 2 occur in the numbers of those houses?



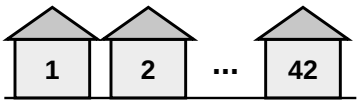
A) 2 **B)** 4 **C)** 8 **D)** 16 **E)** 32

Practice 1.1 Sophia lives on Maple Street where the houses are numbered sequentially from 1 to 35. How many times does the digit 3 occur in the numbers of those houses?



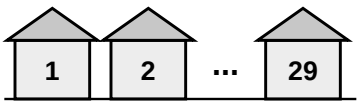
A) 10 **B)** 8 **C)** 9 **D)** 11 **E)** 12

Practice 1.2 A puzzle booklet has its pages numbered sequentially from 1 to 42. How many times does the digit 4 occur in the page numbers?



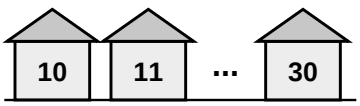
A) 6 **B)** 7 **C)** 8 **D)** 9 **E)** 14

Practice 1.3 The lockers in a school hallway are numbered sequentially from 1 to 29. How many times does the digit 1 occur in the numbers of those lockers?



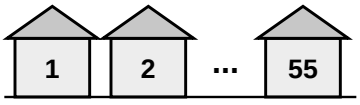
A) 11 **B)** 12 **C)** 14 **D)** 13 **E)** 15

Practice 1.4 The seats in a theater row are numbered sequentially from 10 to 30. How many times does the digit 2 occur in the numbers of those seats?



A) 9 **B)** 10 **C)** 11 **D)** 13 **E)** 12

Practice 1.5 Noah paints house numbers on all the doors along Oak Avenue, numbered sequentially from 1 to 55. How many times does Noah paint the digit 5?



A) 12 **B)** 10 **C)** 11 **D)** 13 **E)** 15

Problem 2: 2002 #11 - Cracked Clock Face

Partitioning the clock face (sum 1 + 2 + ... + 12 = 78) into pieces with given sum properties

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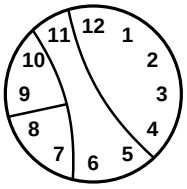
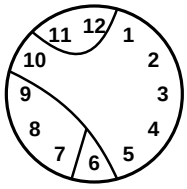
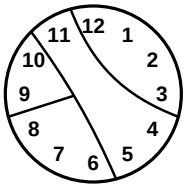
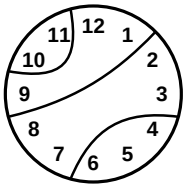
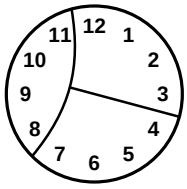
2002 #11 The face of a clock is cracked into 4 pieces. The sums of the numbers within the parts are consecutive numbers. Provided there is only one possible way to crack it, the face would look like:

A) 

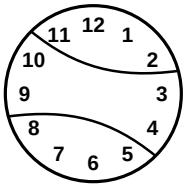
B) 

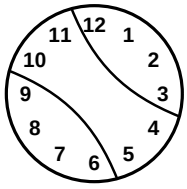
C) 

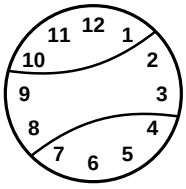
D) 

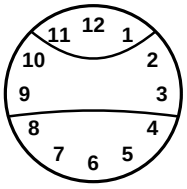
E) 

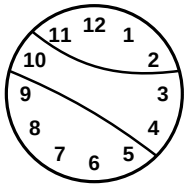
Practice 2.1 The face of a clock is cracked into 3 pieces so that the sums of the numbers on all 3 pieces are equal. Which clock face shows how it was cracked?

A) 

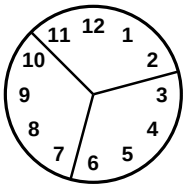
B) 

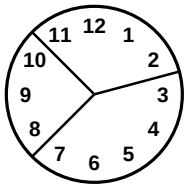
C) 

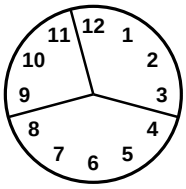
D) 

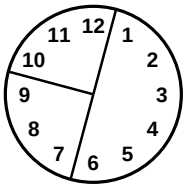
E) 

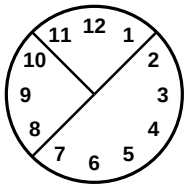
Practice 2.2 The face of a clock is cracked into 3 pieces meeting at the center. The sums of the numbers within the 3 parts are consecutive numbers. Which clock face shows how it was cracked?

A) 

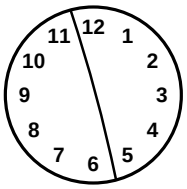
B) 

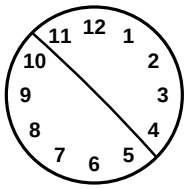
C) 

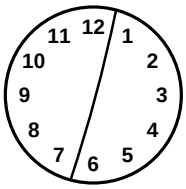
D) 

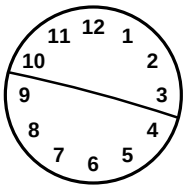
E) 

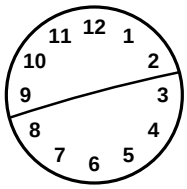
Practice 2.3 A clock face is cracked into 2 pieces by a single crack so that the sum of the numbers on one piece equals the sum of the numbers on the other piece. Which clock face shows this crack?

A) 

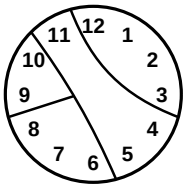
B) 

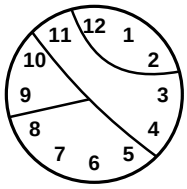
C) 

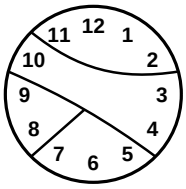
D) 

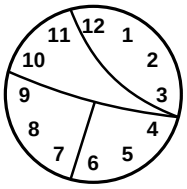
E) 

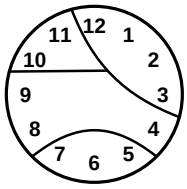
Practice 2.4 The face of a clock is cracked into 4 pieces so that two of the pieces each have a sum of 18, and the other two pieces each have a sum of 21. Which clock face shows this?

A) 

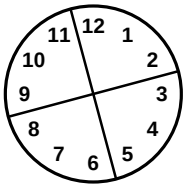
B) 

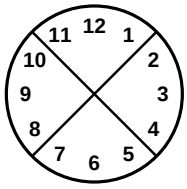
C) 

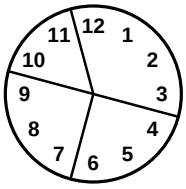
D) 

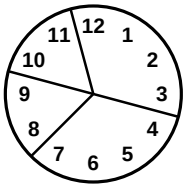
E) 

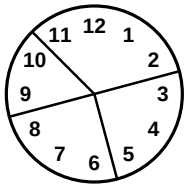
Practice 2.5 The face of a clock is cracked into 4 wedge-shaped pieces meeting at the center. When ordered from smallest to largest, the sums of the pieces increase by 3 each time (15, 18, 21, and 24). Which clock face shows how it was cracked?

A) 

B) 

C) 

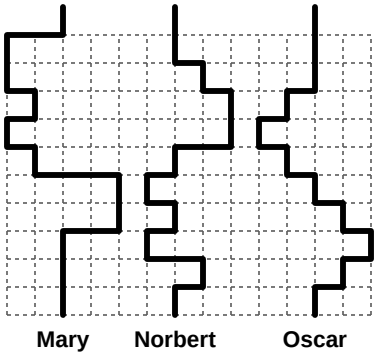
D) 

E) 

Problem 3: 2002 #12 - Kangaroo Zigzag Run

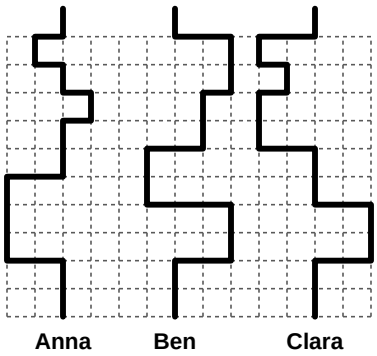
Comparing total grid-segment lengths of zigzag paths

2002 #12 During a zigzag run the kangaroos Mary, Norbert, and Oscar have to jump as drawn in the picture. Suppose they jump at the same speed. Which statement is true?



A) Mary and Oscar arrive at the same time
B) Norbert is first
C) Oscar is last
D) They all arrive at the same time
E) Mary and Norbert arrive at the same time

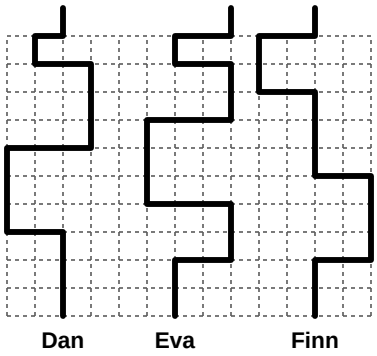
Practice 3.1 During a zigzag run the kangaroos Anna, Ben, and Clara jump along the paths drawn in the picture at the same speed. Which statement is true?



A) Anna is first
B) Ben is first
C) Clara is first
D) Anna is last
E) They all arrive at the same time

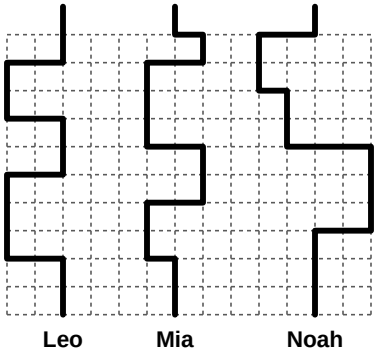
Practice 3.2 The kangaroos Dan, Eva, and Finn jump along the zigzag paths shown at the same speed. Which statement is true?

A) Eva is first
B) Dan and Finn arrive at the same time
C) Finn is last
D) Dan and Eva arrive at the same time
E) They all arrive at the same time



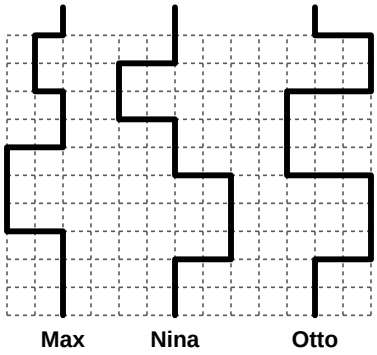
Practice 3.3 Leo, Mia, and Noah jump from the bottom to the top of the grid along the drawn paths at the same speed. Which statement is true?

A) Leo is first
B) Mia is last
C) They all arrive at the same time
D) Only Leo and Mia arrive at the same time
E) Noah is first



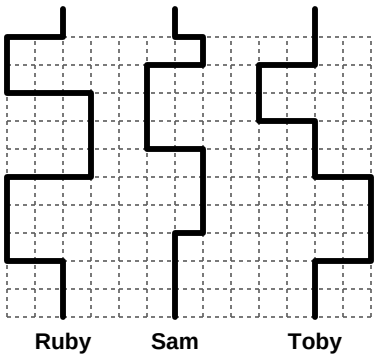
Practice 3.4 Max, Nina, and Otto jump along the paths drawn in the picture at the same speed. Which statement is true?

A) Max is last
B) Nina is first
C) Max and Nina arrive at the same time
D) Otto is last
E) Otto is first



Practice 3.5 Three kangaroos — Ruby, Sam, and Toby — jump from the bottom to the top along the paths shown on the grid, where each small grid square has a side of 1 m. What is the total length of the **shortest** path?

A) 15 m **B)** 16 m **C)** 17 m **D)** 19 m **E)** 21 m



Answer Key & Explanations

For generated/2026-10-19-13-04-28.pdf (Practice set for 2002 #10, 2002 #11, 2002 #12)

Solutions & kid-friendly explanations

2002 #10: C List the house numbers from 1 to 24 that contain the digit 2: 2, 12, 20, 21, 22, 23, and 24. Notice that house number 22 contains two 2s! Counting every digit 2 gives 1 + 1 + 1 + 1 + 2 + 1 + 1 = 8 times.

1.1: A In the ones place, the digit 3 appears in 3, 13, 23, and 33 (4 times). In the tens place, the digit 3 appears in 30, 31, 32, 33, 34, and 35 (6 times). In total, the digit 3 occurs 4 + 6 = 10 times (counting both 3s in 33).

1.2: B In the ones place, the digit 4 appears in 4, 14, 24, and 34 (4 times). In the tens place, the digit 4 appears in 40, 41, and 42 (3 times). In total, the digit 4 occurs 4 + 3 = 7 times.

1.3: D In the ones place, the digit 1 appears in 1, 11, and 21 (3 times). In the tens place, the digit 1 appears in all ten numbers from 10 to 19 (10 times). In total, the digit 1 occurs 3 + 10 = 13 times (with 11 contributing two 1s).

1.4: E Between 10 and 30, the digit 2 appears in the ones place in 12 and 22 (2 times), and in the tens place in every number from 20 to 29 (10 times). In total, the digit 2 occurs 2 + 10 = 12 times.

1.5: A In the ones place, the digit 5 appears in 5, 15, 25, 35, 45, and 55 (6 times). In the tens place, the digit 5 appears in 50, 51, 52, 53, 54, and 55 (6 times). Altogether, Noah paints the digit 5 a total of 6 + 6 = 12 times.

2002 #11: C The sum of all numbers on a clock face is 1 + 2 + ... + 12 = 78. Four consecutive numbers that add up to 78 must be 18, 19, 20, and 21 (since 18 + 19 + 20 + 21 = 78). In clock C, the four pieces contain {12, 1, 2, 3} = 18, {9, 10} = 19, {11, 4, 5} = 20, and {6, 7, 8} = 21.

2.1: A Since the numbers 1 through 12 add up to 78, each of the 3 equal pieces must have a sum of 78 ÷ 3 = 26. In clock A, the top piece has 11 + 12 + 1 + 2 = 26, the middle piece has 9 + 10 + 3 + 4 = 26, and the bottom piece has 5 + 6 + 7 + 8 = 26.

2.2: B Three consecutive numbers that add up to 78 are 25, 26, and 27 (25 + 26 + 27 = 78). In clock B, the three pieces are {3, 4, 5, 6, 7} = 25, {11, 12, 1, 2} = 26, and {8, 9, 10} = 27.

2.3: D Half of the total clock sum is 78 ÷ 2 = 39. In clock D, the crack goes between 9 and 10 on the left and between 3 and 4 on the right, giving {10, 11, 12, 1, 2, 3} = 39 and {4, 5, 6, 7, 8, 9} = 39.

2.4: E In clock E, the top-right piece has {12, 1, 2, 3} = 18, the bottom piece has {5, 6, 7} = 18, the top-left piece has {10, 11} = 21, and the middle piece has {8, 9, 4} = 21.

2.5: C In clock C, the four wedges are {4, 5, 6} = 15, {12, 1, 2, 3} = 18, {10, 11} = 21, and {7, 8, 9} = 24. Sorted in order, the sums are 15, 18, 21, and 24.

2002 #12: E All three kangaroos move upward by the exact same vertical distance (11 grid units) without ever going down. Counting their horizontal side-to-side steps: Mary makes 2 + 3 + 1 + 1 + 1 + 2 = 10 horizontal steps (total 21), Norbert makes 1 + 2 + 1 + 1 + 1 + 2 + 1 + 1 = 10 horizontal steps (total 21), and Oscar makes 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8 horizontal steps (total 19). Oscar finishes first (19 steps), while Mary and Norbert arrive at the same time (21 steps each).

3.1: A Each kangaroo travels 11 units upward. Counting horizontal steps: Anna takes 2 + 2 + 1 + 1 + 1 + 1 = 8 horizontal steps (total 19), Ben takes 2 + 3 + 2 + 1 + 2 = 10 horizontal steps (total 21), and Clara takes 2 + 2 + 2 + 1 + 1 + 2 = 10 horizontal steps (total 21). Since 19 is the shortest path, Anna finishes first.

3.2: B All three paths go 11 units upward. Counting horizontal steps: Dan takes 2 + 3 + 2 + 1 = 8 horizontal steps (total 19), Eva takes 2 + 3 + 3 + 2 + 2 = 12 horizontal steps (total 23), and Finn takes 2 + 2 + 2 + 2 = 8 horizontal steps (total 19). Therefore, Dan and Finn arrive at the same time.

3.3: C Every kangaroo moves 11 units upward. Counting horizontal steps: Leo takes 2 + 2 + 2 + 2 = 8 horizontal steps (total 19), Mia takes 1 + 2 + 2 + 2 + 1 = 8 horizontal steps (total 19), and Noah takes 2 + 3 + 1 + 2 = 8 horizontal steps (total 19). Since all three travel 19 grid segments, they all arrive at the same time.

3.4: D All three paths have 11 vertical steps upward. Counting horizontal steps: Max takes 2 + 2 + 1 + 1 = 6 horizontal steps (total 17), Nina takes 2 + 2 + 2 + 2 = 8 horizontal steps (total 19), and Otto takes 2 + 3 + 3 + 2 = 10 horizontal steps (total 21). Otto has the longest path (21 steps), so Otto is last.

3.5: C Every kangaroo travels 11 m upward. Counting horizontal steps: Ruby travels 2 + 3 + 3 + 2 = 10 m horizontally (total 11 + 10 = 21 m), Sam travels 1 + 2 + 2 + 1 = 6 m horizontally (total 11 + 6 = 17 m), and Toby travels 2 + 2 + 2 + 2 = 8 m horizontally (total 11 + 8 = 19 m). The shortest path is Sam's path at 17 m.